

Bragg Method Data analysis

Plotting graphs for peak identification in Bragg method, as well as other stuff.

```
In [3]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

Graph Visualisation NaCl (100) first measurement:

```
In [5]: # Read into pandas DataFrame
NaCl_100df = pd.read_csv(
    r"C:\Users\joshb\OneDrive\PHYSICS BSc UNIVERSITY of EXETER\BSc Physics Stage 2\Practical Physics II\practical_physics_II\Lab_2_Bragg_La
    sep="\t",
    encoding="latin1"
)
```

```
In [6]: # Rename columns
NaCl_100df.columns = ["theta_deg", "count_rate"]
```

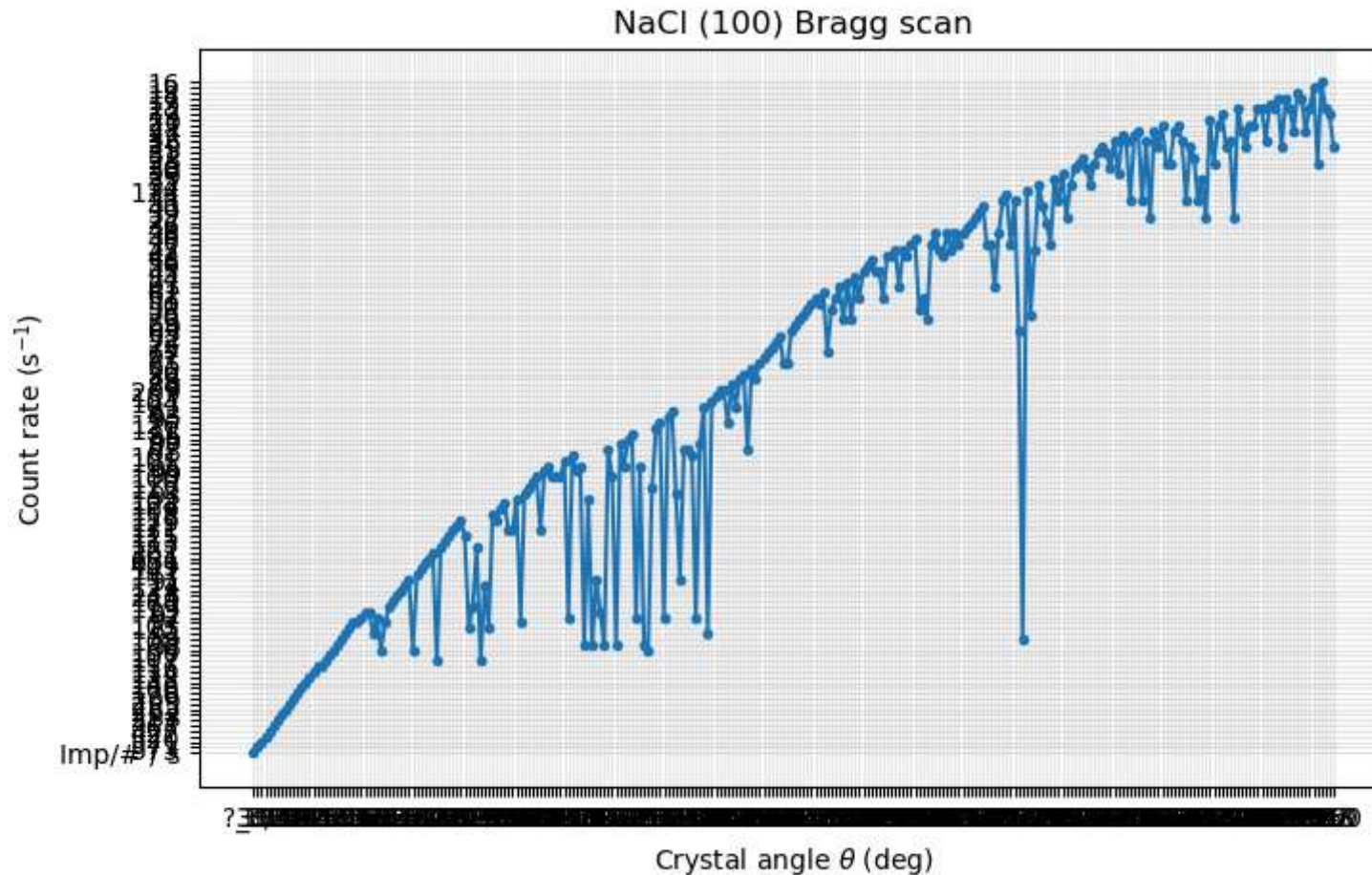
```
In [7]: display(NaCl_100df.head())
print("Data shape:", NaCl_100df.shape)
```

	theta_deg	count_rate
0	?_c/°	Imp/# / s
1	3.0	573
2	3.1	571
3	3.2	520
4	3.3	422

Data shape: (272, 2)

```
In [8]: plt.figure(figsize=(8,5))
plt.plot(NaCl_100df["theta_deg"], NaCl_100df["count_rate"], 'o-', markersize=3)
plt.xlabel(r"Crystal angle $\theta$ (deg)")
plt.ylabel("Count rate (s$^{-1}$)")
```

```
plt.title("NaCl (100) Bragg scan")
plt.grid(alpha=0.3)
plt.show()
```



^ Ok, this looks like nonsense. The main issue is row 0 is not data. It is the second header line from the Measure export, which pandas has tried to treat as a data row.

This means that theta_deg is an object column, not float. This means matplotlib is treating theta as categorical, causing the weird plots.

```
In [10]: # Drop the first row
NaCl_100df = NaCl_100df.drop(index=0).reset_index(drop=True)
```

```
In [11]: # Convert data type of both columns to numeric
NaCl_100df["theta_deg"] = pd.to_numeric(NaCl_100df["theta_deg"])
NaCl_100df["count_rate"] = pd.to_numeric(NaCl_100df["count_rate"])
```

```
In [12]: # Sanity check
NaCl_100df.dtypes
NaCl_100df.head()
```

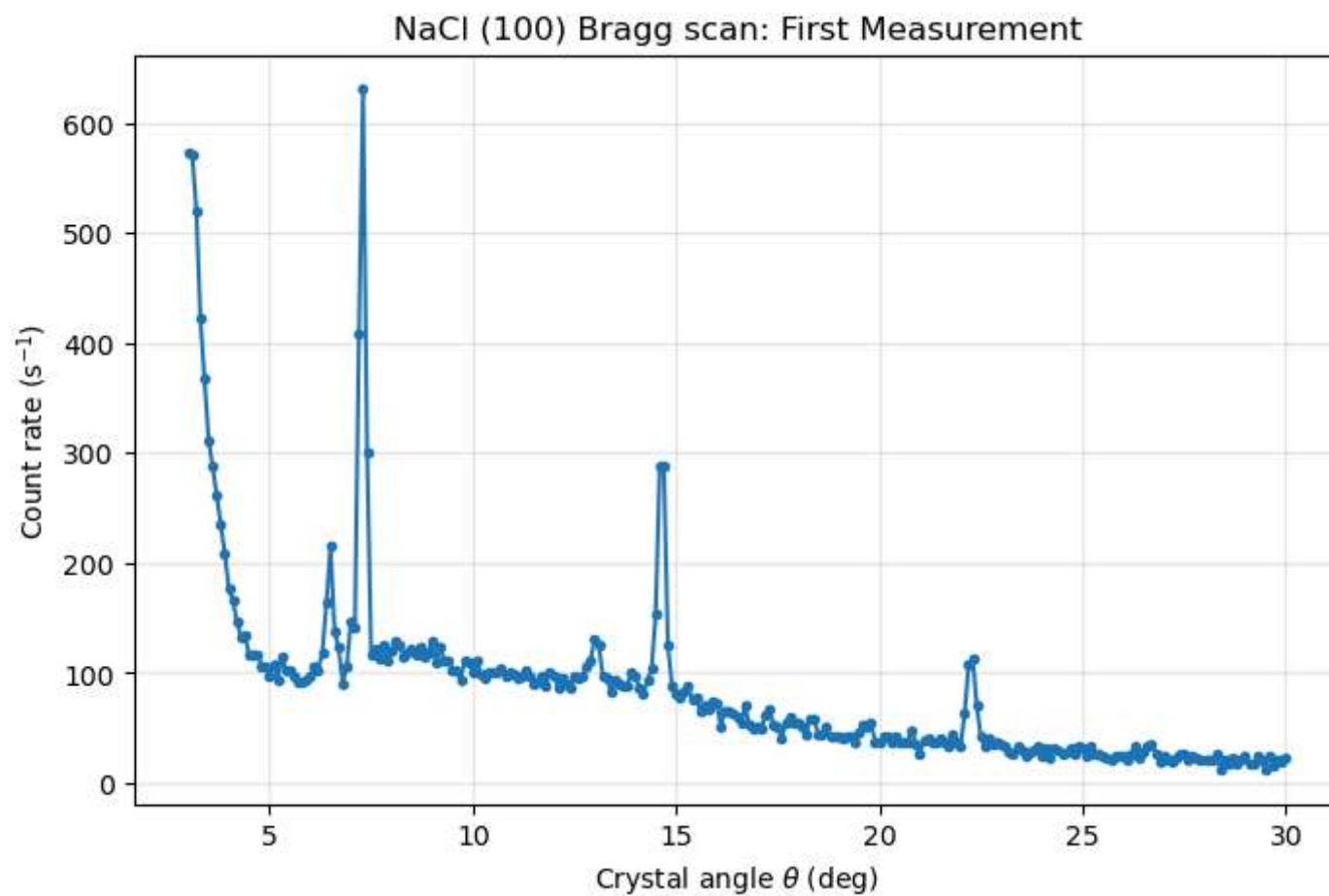
```
Out[12]:
```

	theta_deg	count_rate
0	3.0	573
1	3.1	571
2	3.2	520
3	3.3	422
4	3.4	367

```
In [13]: plt.figure(figsize=(8,5))
plt.plot(NaCl_100df["theta_deg"], NaCl_100df["count_rate"], 'o-', markersize=3)
plt.xlabel(r"Crystal angle $\theta$ (deg)")
plt.ylabel("Count rate (s$^{-1}$)")
plt.title("NaCl (100) Bragg scan: First Measurement")
plt.grid(alpha=0.3)

# Save to specific location
plt.savefig(
    r"C:\Users\joshb\OneDrive\PHYSICS BSc UNIVERSITY of EXETER\BSc Physics Stage 2\Practical Physics II\practical_physics_II\Lab_2_Bragg_La
    dpi=300,
    bbox_inches="tight"
)

plt.show()
```



Graph Visualisation NaCl (100) second measurement:

```
In [15]: # Read into pandas DataFrame
NaCl_100df_b = pd.read_csv(
    r"C:\Users\joshb\OneDrive\PHYSICS BSc UNIVERSITY of EXETER\BSc Physics Stage 2\Practical Physics II\practical_physics_II\Lab_2_Bragg_La
    sep="\t",
    encoding="latin1"
)
```

```
In [16]: # Rename columns
NaCl_100df_b.columns = ["theta_deg", "count_rate"]
```

```
In [17]: display(NaCl_100df_b.head())
print("Data shape:", NaCl_100df_b.shape)
```

	theta_deg	count_rate
0	3.0	210
1	3.1	217
2	3.2	244
3	3.3	227
4	3.4	195

Data shape: (271, 2)

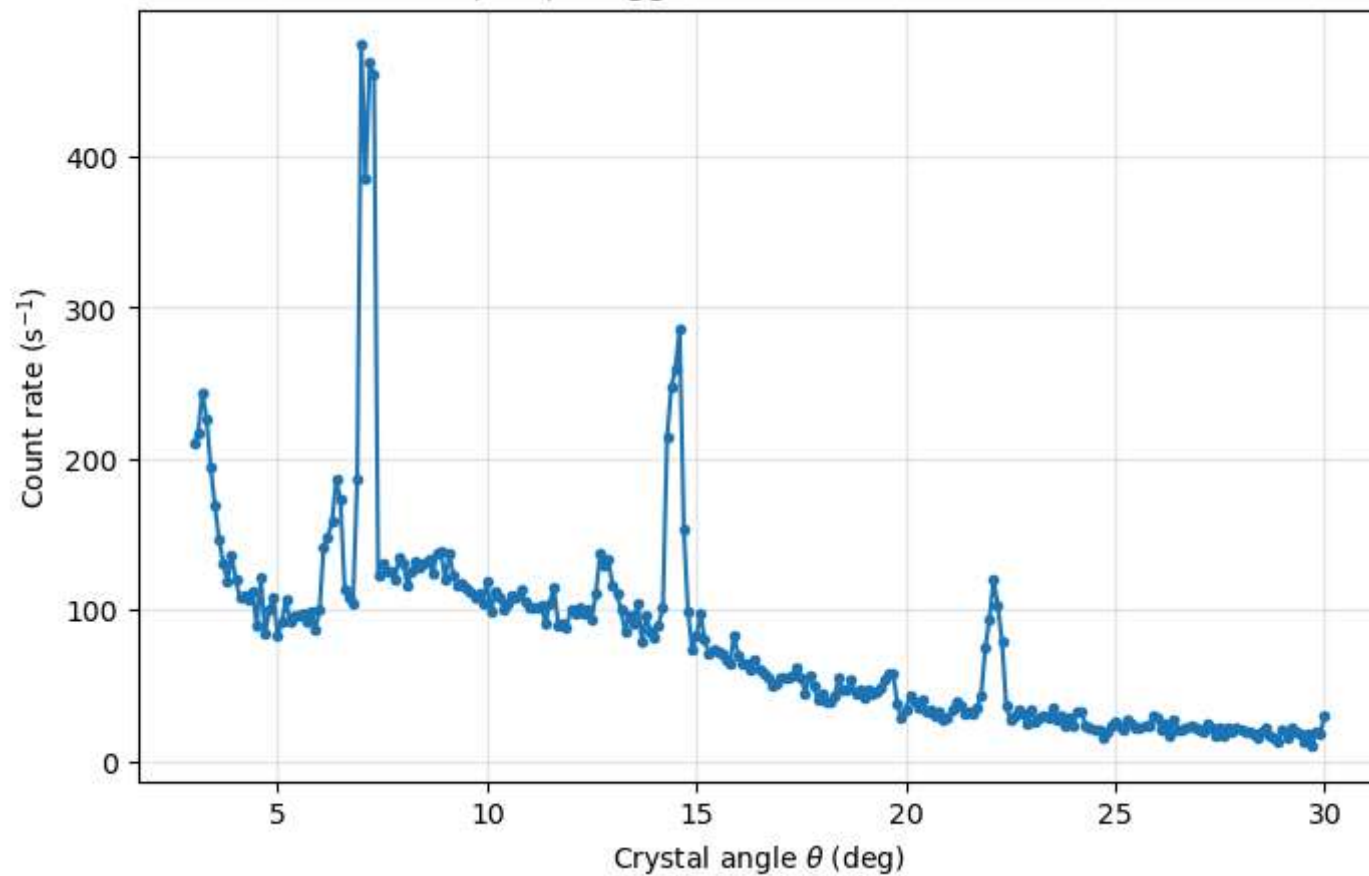
The first row is numerical, so no need to drop first row and change data type. Data shape confirms there is no first row (one fewer than previous).

```
In [19]: plt.figure(figsize=(8,5))
plt.plot(NaCl_100df_b["theta_deg"], NaCl_100df_b["count_rate"], 'o-', markersize=3)
plt.xlabel(r"Crystal angle $\theta$ (deg)")
plt.ylabel("Count rate (s$^{-1}$)")
plt.title("NaCl (100) Bragg scan: Second Measurement")
plt.grid(alpha=0.3)

# Save to specific location
plt.savefig(
    r"C:\Users\joshb\OneDrive\PHYSICS BSc UNIVERSITY of EXETER\BSc Physics Stage 2\Practical Physics II\practical_physics_II\Lab_2_Bragg_La
    dpi=300,
    bbox_inches="tight"
)

plt.show()
```

NaCl (100) Bragg scan: Second Measurement



Graph Visualisation NaCl (110) first measurement:

```
In [21]: # Read into pandas DataFrame
NaCl_110df = pd.read_csv(
    r"C:\Users\joshb\OneDrive\PHYSICS BSc UNIVERSITY of EXETER\BSc Physics Stage 2\Practical Physics II\practical_physics_II\Lab_2_Bragg_La
    sep="\t",
    encoding="latin1"
)
```

```
In [22]: # Rename columns
NaCl_110df.columns = ["theta_deg", "count_rate"]
```

```
In [23]: display(NaCl_110df.head())
print("Data shape:", NaCl_110df.shape)
```

	theta_deg	count_rate
0	?_c/°	Imp/# / s
1	3.0	444
2	3.1	379
3	3.2	370
4	3.3	299

Data shape: (272, 2)

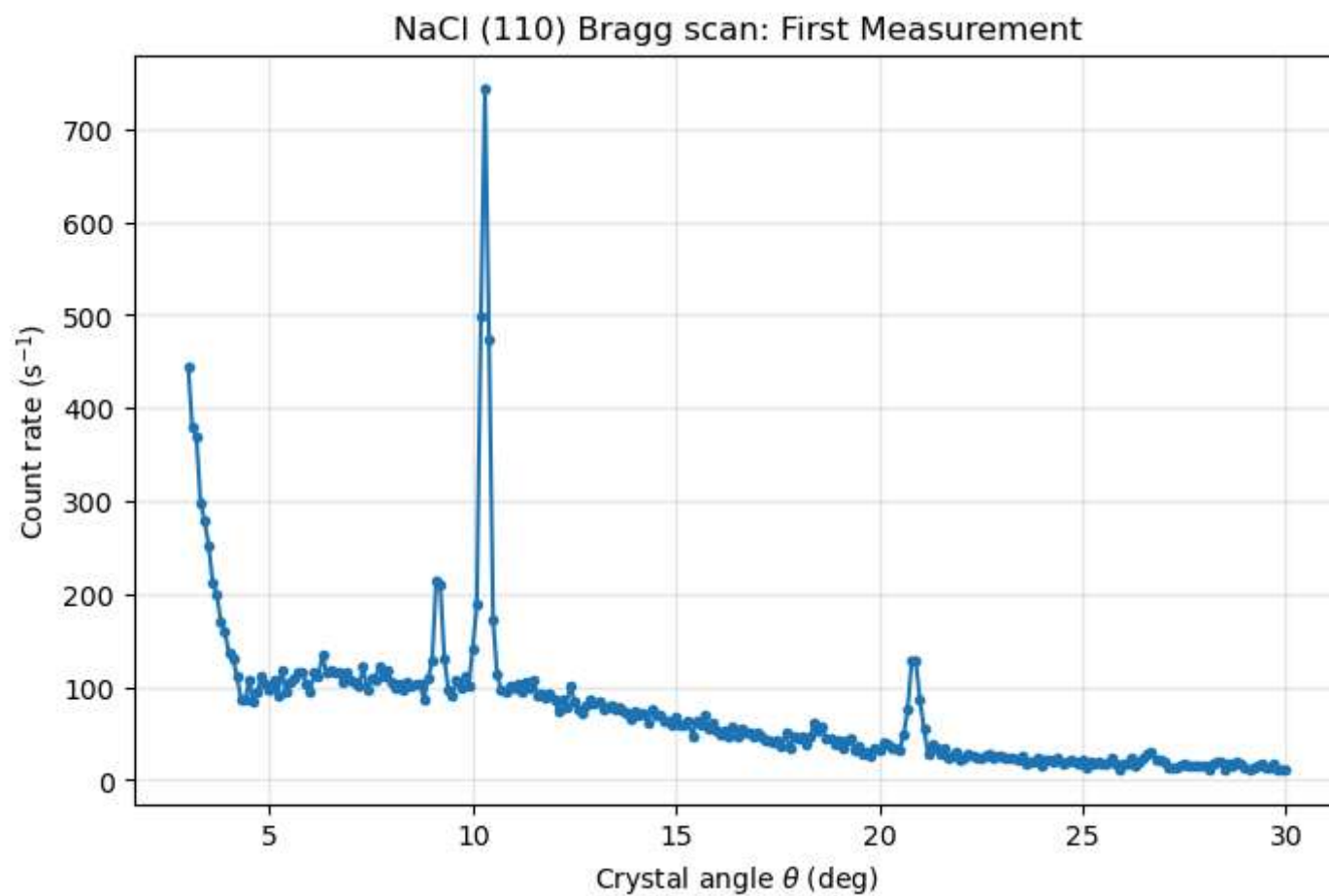
```
In [24]: # Drop the first row
NaCl_110df = NaCl_110df.drop(index=0).reset_index(drop=True)
```

```
In [25]: # Convert data type of both columns to numeric
NaCl_110df["theta_deg"] = pd.to_numeric(NaCl_110df["theta_deg"])
NaCl_110df["count_rate"] = pd.to_numeric(NaCl_110df["count_rate"])
```

```
In [26]: plt.figure(figsize=(8,5))
plt.plot(NaCl_110df["theta_deg"], NaCl_110df["count_rate"], 'o-', markersize=3)
plt.xlabel(r"Crystal angle $\theta$ (deg)")
plt.ylabel("Count rate (s$^{-1}$)")
plt.title("NaCl (110) Bragg scan: First Measurement")
plt.grid(alpha=0.3)

# Save to specific location
plt.savefig(
    r"C:\Users\joshb\OneDrive\PHYSICS BSc UNIVERSITY of EXETER\BSc Physics Stage 2\Practical Physics II\practical_physics_II\Lab_2_Bragg_Laue\NaCl_110_Bragg_scan.png",
    dpi=300,
    bbox_inches="tight"
)

plt.show()
```



Graph Visualisation NaCl (110) second measurement:

```
In [28]: # Read into pandas DataFrame
NaCl_110df_b = pd.read_csv(
    r"C:\Users\joshb\OneDrive\PHYSICS BSc UNIVERSITY of EXETER\BSc Physics Stage 2\Practical Physics II\practical_physics_II\Lab_2_Bragg_La
    sep="\t",
    encoding="latin1"
)

In [29]: # Rename columns
NaCl_110df_b.columns = ["theta_deg", "count_rate"]

In [30]: display(NaCl_110df_b.head())
print("Data shape:", NaCl_110df_b.shape)
```


	theta_deg	count_rate
0	3.0	321
1	3.1	334
2	3.2	301
3	3.3	320
4	3.4	266

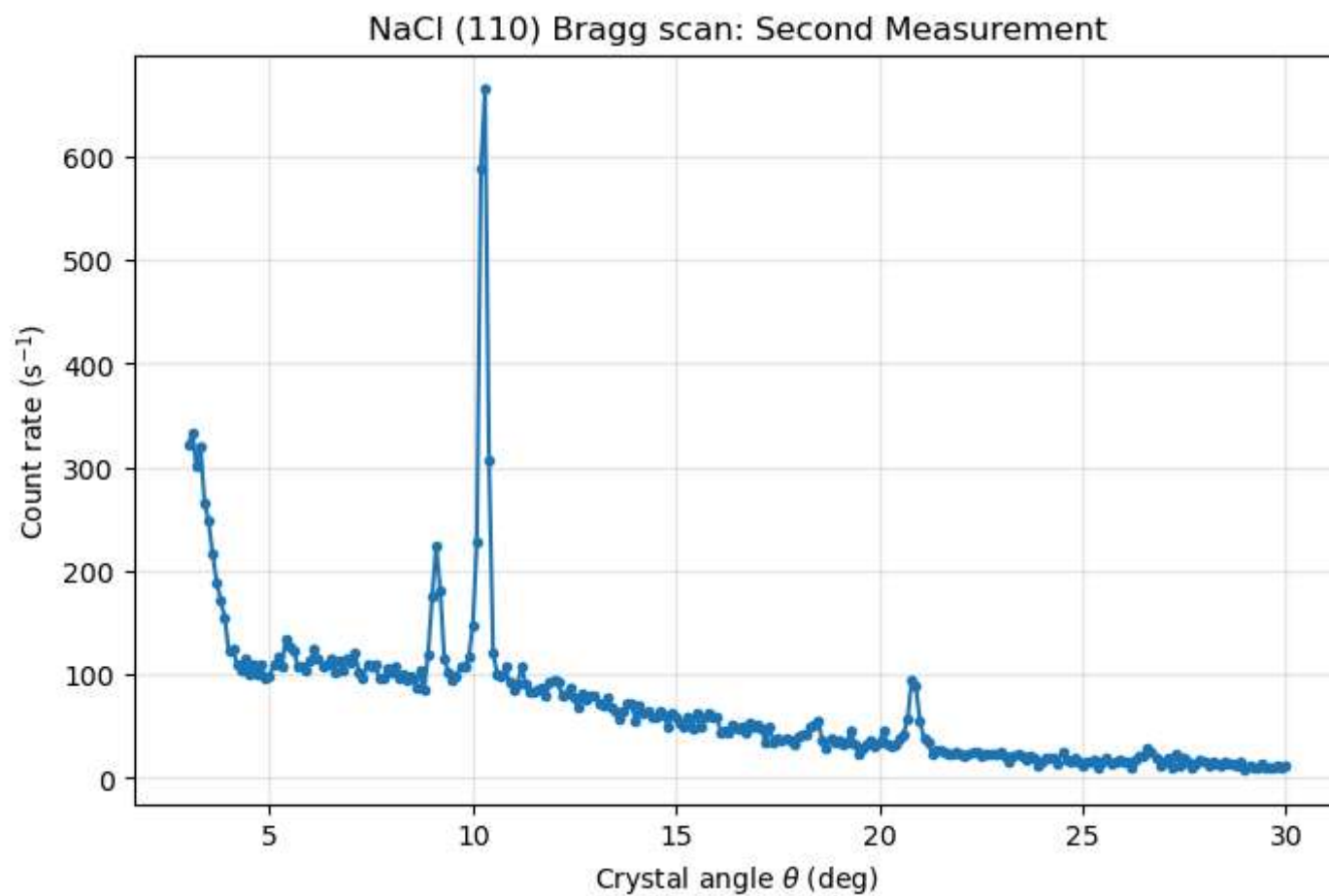
Data shape: (271, 2)

The first row is numerical, so no need to drop first row and change data type. Data shape confirms there is no first row (one fewer than previous).

```
In [32]: plt.figure(figsize=(8,5))
plt.plot(NaCl_110df_b["theta_deg"], NaCl_110df_b["count_rate"], 'o-', markersize=3)
plt.xlabel(r"Crystal angle $\theta$ (deg)")
plt.ylabel("Count rate (s$^{-1}$)")
plt.title("NaCl (110) Bragg scan: Second Measurement")
plt.grid(alpha=0.3)

# Save to specific location
plt.savefig(
    r"C:\Users\joshb\OneDrive\PHYSICS BSc UNIVERSITY of EXETER\BSc Physics Stage 2\Practical Physics II\practical_physics_II\Lab_2_Bragg_La
    dpi=300,
    bbox_inches="tight"
)

plt.show()
```



Graph Visualisation NaCl (111) first measurement:

```
In [34]: # Read into pandas DataFrame
NaCl_111df = pd.read_csv(
    r"C:\Users\joshb\OneDrive\PHYSICS BSc UNIVERSITY of EXETER\BSc Physics Stage 2\Practical Physics II\practical_physics_II\Lab_2_Bragg_La
    sep="\t",
    encoding="latin1"
)

In [35]: # Rename columns
NaCl_111df.columns = ["theta_deg", "count_rate"]

In [36]: display(NaCl_111df.head())
print("Data shape:", NaCl_111df.shape)
```

	theta_deg	count_rate
0	3.0	7319
1	3.1	6614
2	3.2	4899
3	3.3	3056
4	3.4	1660

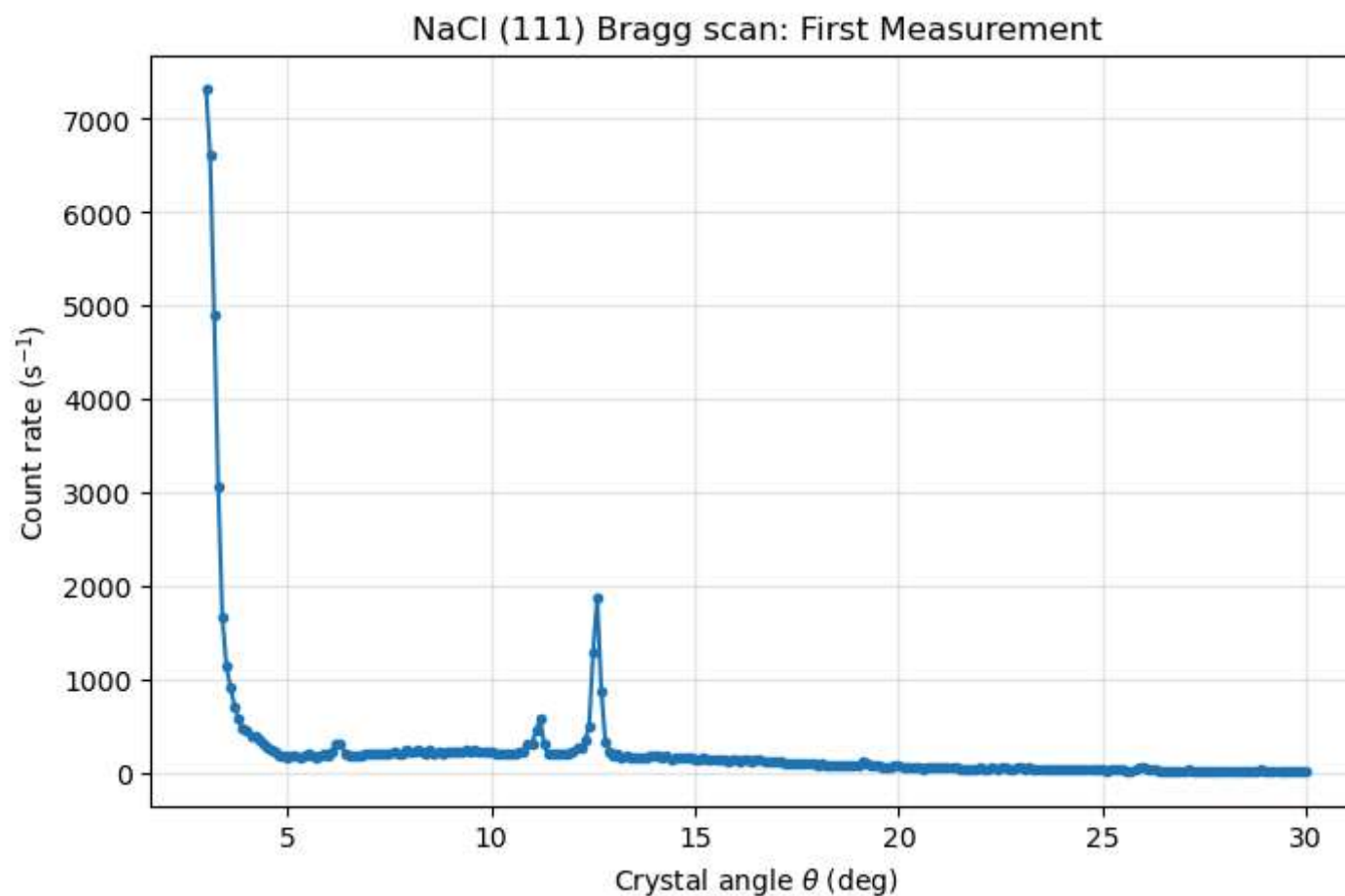
Data shape: (271, 2)

No need to change data or drop first row.

```
In [38]: plt.figure(figsize=(8,5))
plt.plot(NaCl_111df["theta_deg"], NaCl_111df["count_rate"], 'o-', markersize=3)
plt.xlabel(r"Crystal angle $\theta$ (deg)")
plt.ylabel("Count rate (s$^{-1}$)")
plt.title("NaCl (111) Bragg scan: First Measurement")
plt.grid(alpha=0.3)

# Save to specific location
plt.savefig(
    r"C:\Users\joshb\OneDrive\PHYSICS BSc UNIVERSITY of EXETER\BSc Physics Stage 2\Practical Physics II\practical_physics_II\Lab_2_Bragg_La
    dpi=300,
    bbox_inches="tight"
)

plt.show()
```



This plot is dominated by high count rates at low crystal angles $0^\circ \leq \theta \leq 4^\circ$, making identification of higher-angle Bragg peaks difficult on the same scale.

The elevated count rate at low angles is consistent with the straight-through X-ray beam and/or an angular zero offset in the crystal alignment. Such behaviour is commonly observed when the crystal zero position is imperfectly calibrated, resulting in a constant angular shift across the scan.

A constant angular offset shifts all peak positions by approximately the same amount but does not affect the relative angular separation or ratios between peaks. As a result, qualitative identification of allowed reflections and comparison of peak structure remain valid, while absolute Bragg angles extracted from this dataset are subject to increased systematic uncertainty.

To improve visual identification of Bragg peaks, subsequent plots are rescaled to exclude the low-angle region dominating the intensity scale. The full dataset is retained, and the potential angular offset is treated explicitly as a source of systematic uncertainty in the analysis.

```
In [40]: theta_min = 5.0 # degrees
```

```

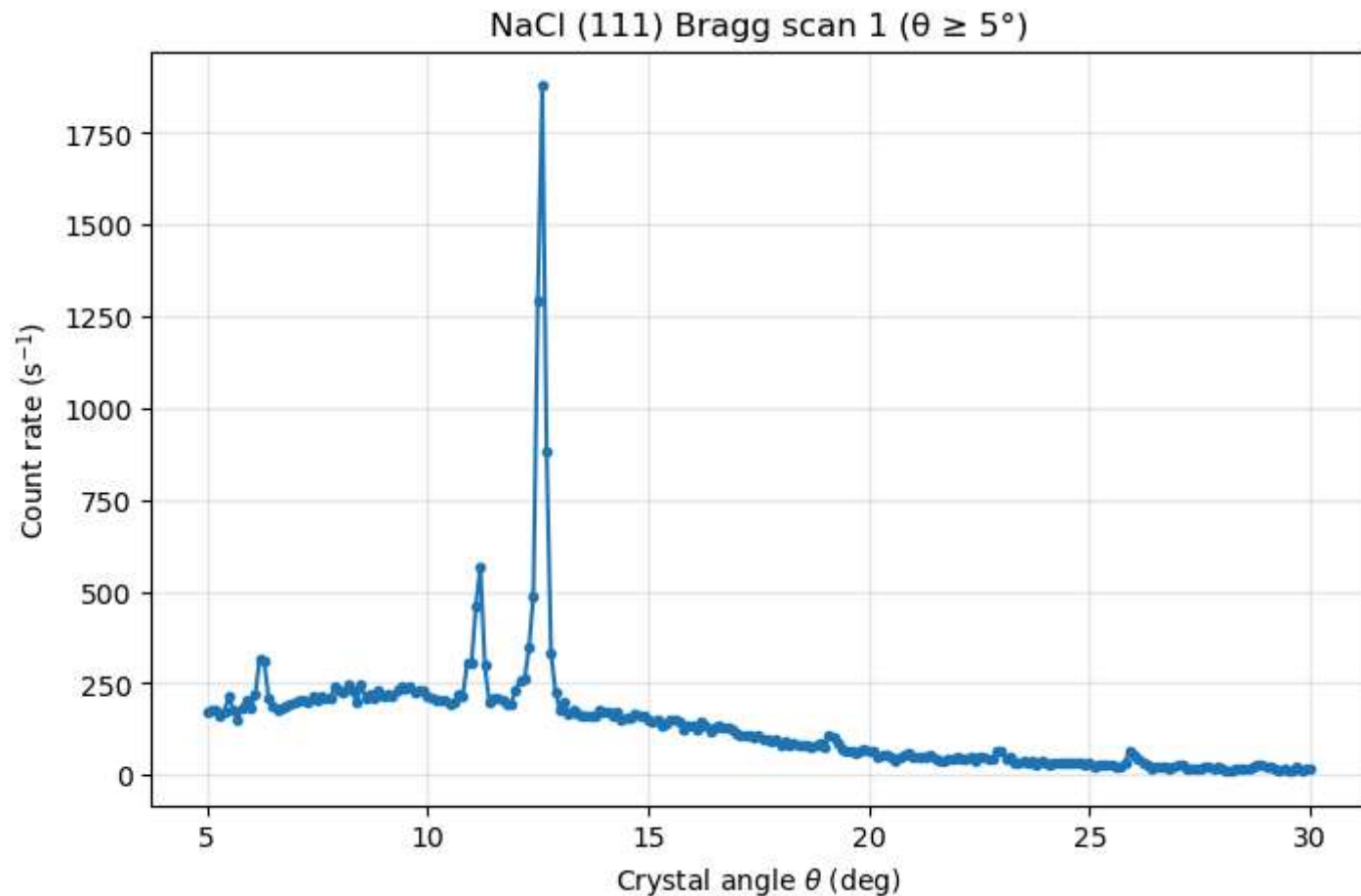
roi = NaCl_111df[NaCl_111df["theta_deg"] >= theta_min]

plt.figure(figsize=(8,5))
plt.plot(roi["theta_deg"], roi["count_rate"], 'o-', markersize=3)
plt.xlabel(r"Crystal angle $\theta$ (deg)")
plt.ylabel(r"Count rate (s$^{-1}$)")
plt.title("NaCl (111) Bragg scan 1 ( $\theta \geq 5^\circ$ ")
plt.grid(alpha=0.3)

# Save to specific location
plt.savefig(
    r"C:\Users\joshb\OneDrive\PHYSICS BSc UNIVERSITY of EXETER\BSc Physics Stage 2\Practical Physics II\practical_physics_II\Lab_2_Bragg_La
    dpi=300,
    bbox_inches="tight"
)

plt.show()

```



Graph Visualisation NaCl (111) second measurement:

```
In [42]: # Read into pandas DataFrame
NaCl_111df_b = pd.read_csv(
    r"C:\Users\joshb\OneDrive\PHYSICS BSc UNIVERSITY of EXETER\BSc Physics Stage 2\Practical Physics II\practical_physics_II\Lab_2_Bragg_La
    sep="\t",
    encoding="latin1"
)
```

```
In [43]: # Rename columns
NaCl_111df_b.columns = ["theta_deg", "count_rate"]
```

```
In [44]: display(NaCl_111df.head())
print("Data shape:", NaCl_111df.shape)
```

	theta_deg	count_rate
0	3.0	7319
1	3.1	6614
2	3.2	4899
3	3.3	3056
4	3.4	1660

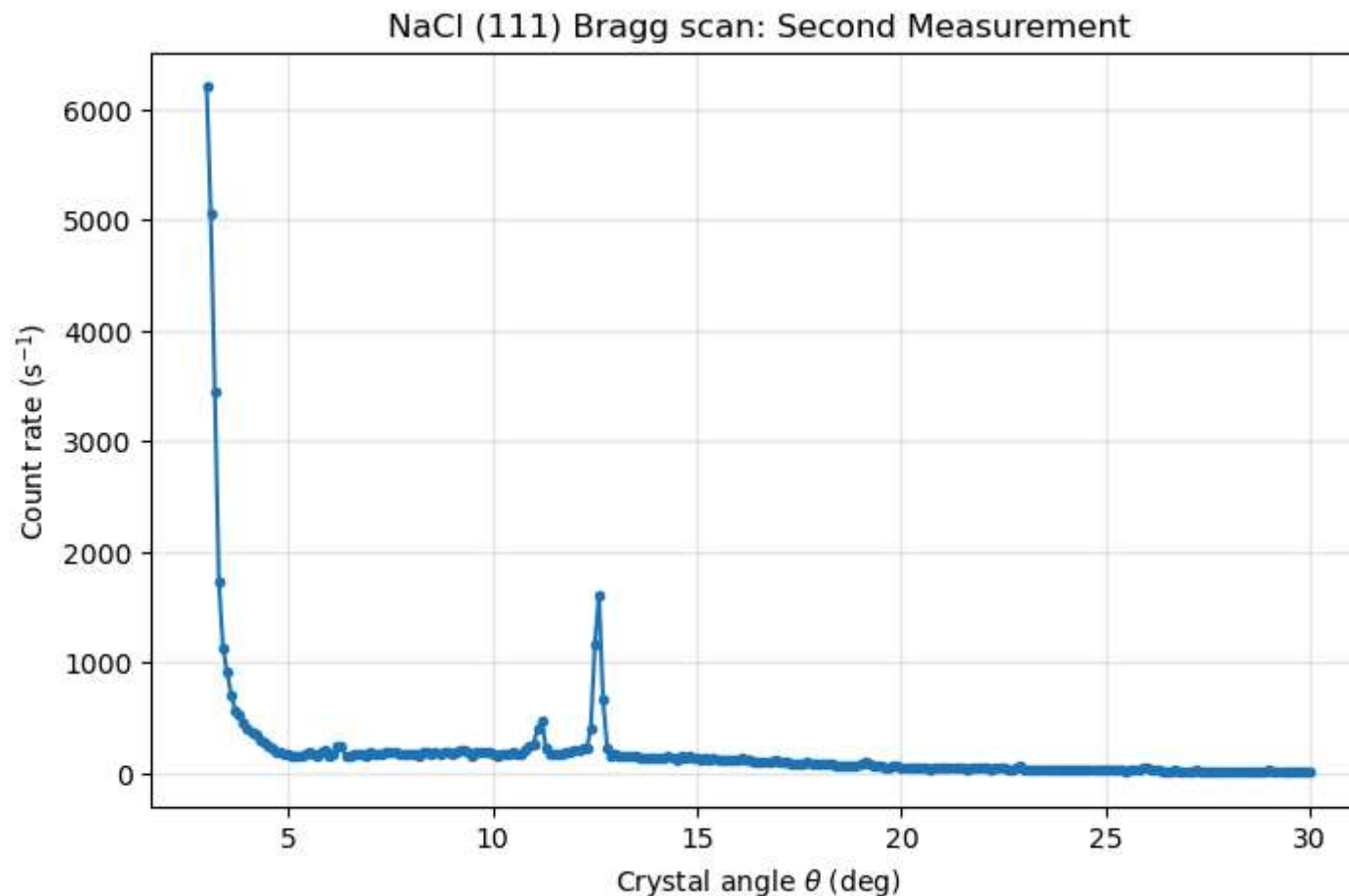
Data shape: (271, 2)

Code numeric, don't need to drop first row.

```
In [46]: plt.figure(figsize=(8,5))
plt.plot(NaCl_111df_b["theta_deg"], NaCl_111df_b["count_rate"], 'o-', markersize=3)
plt.xlabel(r"Crystal angle $\theta$ (deg)")
plt.ylabel("Count rate (s$^{-1}$)")
plt.title("NaCl (111) Bragg scan: Second Measurement")
plt.grid(alpha=0.3)

# Save to specific location
plt.savefig(
    r"C:\Users\joshb\OneDrive\PHYSICS BSc UNIVERSITY of EXETER\BSc Physics Stage 2\Practical Physics II\practical_physics_II\Lab_2_Bragg_La
    dpi=300,
    bbox_inches="tight"
)
```

```
plt.show()
```



Again, dominated by initial high readings. I will perform same offset plotting to be able to identify peaks.

I am now thinking that it was an issue with calibration, which supports my earlier hypothesis that the machine held the crystal at too low an angle initially and thus a straight through reading was picked up with the X-Rays firing straight onto the tube. If calibration was off, then zero positions of detector and crystal would be incorrect. It could even have been that I forgot to recalibrate the X-ray unit before taking NaCl (111) readings.

```
In [48]: # Using same value for theta_min. The fact that it stops around the same point more and more makes me think
# it is a calibration issue that has introduced a zero error in degrees
```

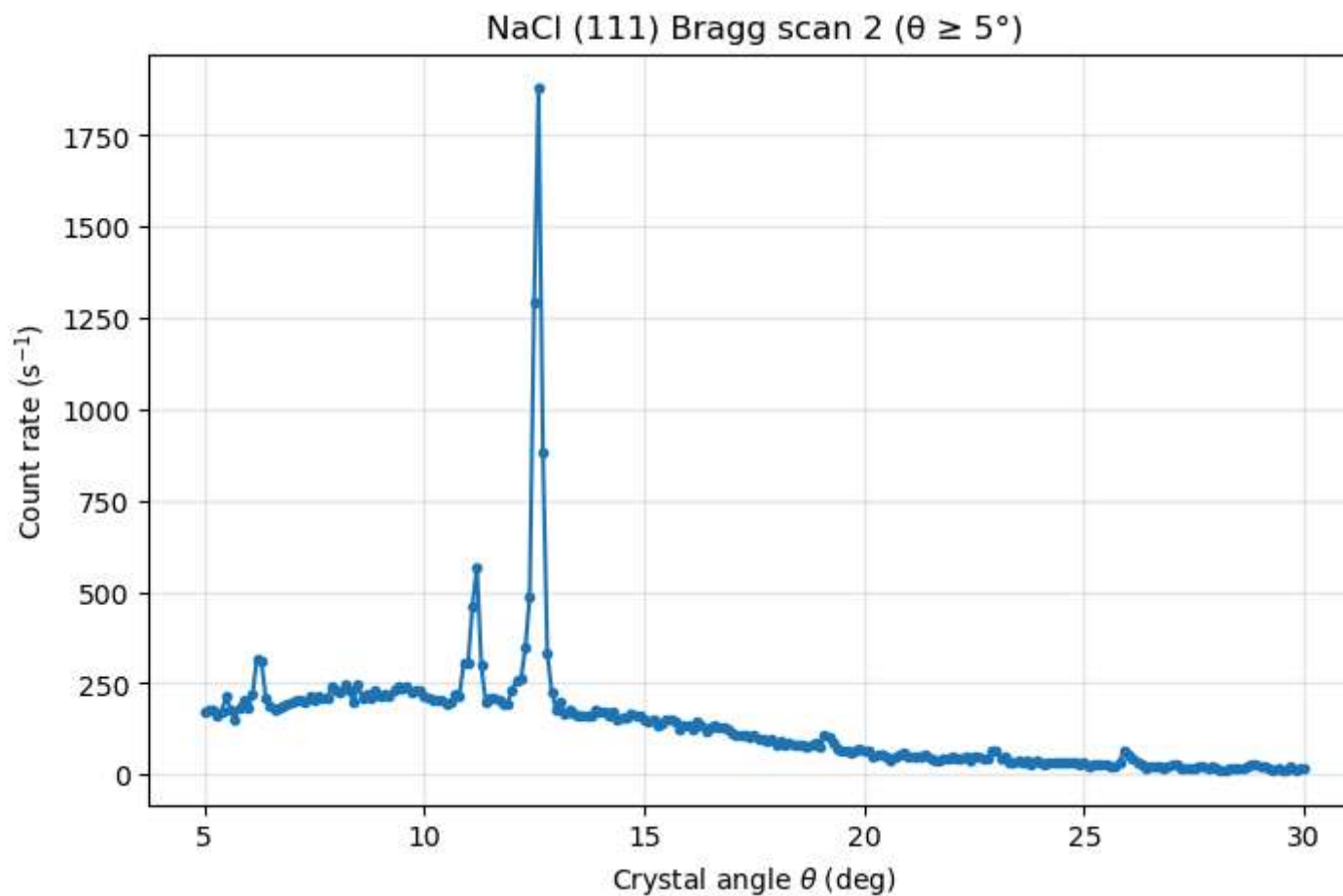
```
roi2 = NaCl_111df[NaCl_111df["theta_deg"] >= theta_min]
```

```
plt.figure(figsize=(8,5))
plt.plot(roi2["theta_deg"], roi2["count_rate"], 'o-', markersize=3)
plt.xlabel(r"Crystal angle  $\theta$  (deg)")
```

```
plt.ylabel(r"Count rate (s-1)")
plt.title("NaCl (111) Bragg scan 2 ( $\theta \geq 5^\circ$ ")
plt.grid(alpha=0.3)

# Save to specific location
plt.savefig(
    r"C:\Users\joshb\OneDrive\PHYSICS BSc UNIVERSITY of EXETER\BSc Physics Stage 2\Practical Physics II\practical_physics_II\Lab_2_Bragg_Laue",
    dpi=300,
    bbox_inches="tight"
)

plt.show()
```



Graph Visualisation LiF (100) measurement:


```
In [50]: # Read into pandas DataFrame
LiF_100df = pd.read_csv(
    r"C:\Users\joshb\OneDrive\PHYSICS BSc UNIVERSITY of EXETER\BSc Physics Stage 2\Practical Physics II\practical_physics_II\Lab_2_Bragg_La
    sep="\t",
    encoding="latin1"
)
```

```
In [51]: # Rename columns
LiF_100df.columns = ["theta_deg", "count_rate"]
```

```
In [52]: display(LiF_100df.head())
print("Data shape:", LiF_100df.shape)
```

	theta_deg	count_rate
0	3.0	270
1	3.1	254
2	3.2	217
3	3.3	190
4	3.4	158

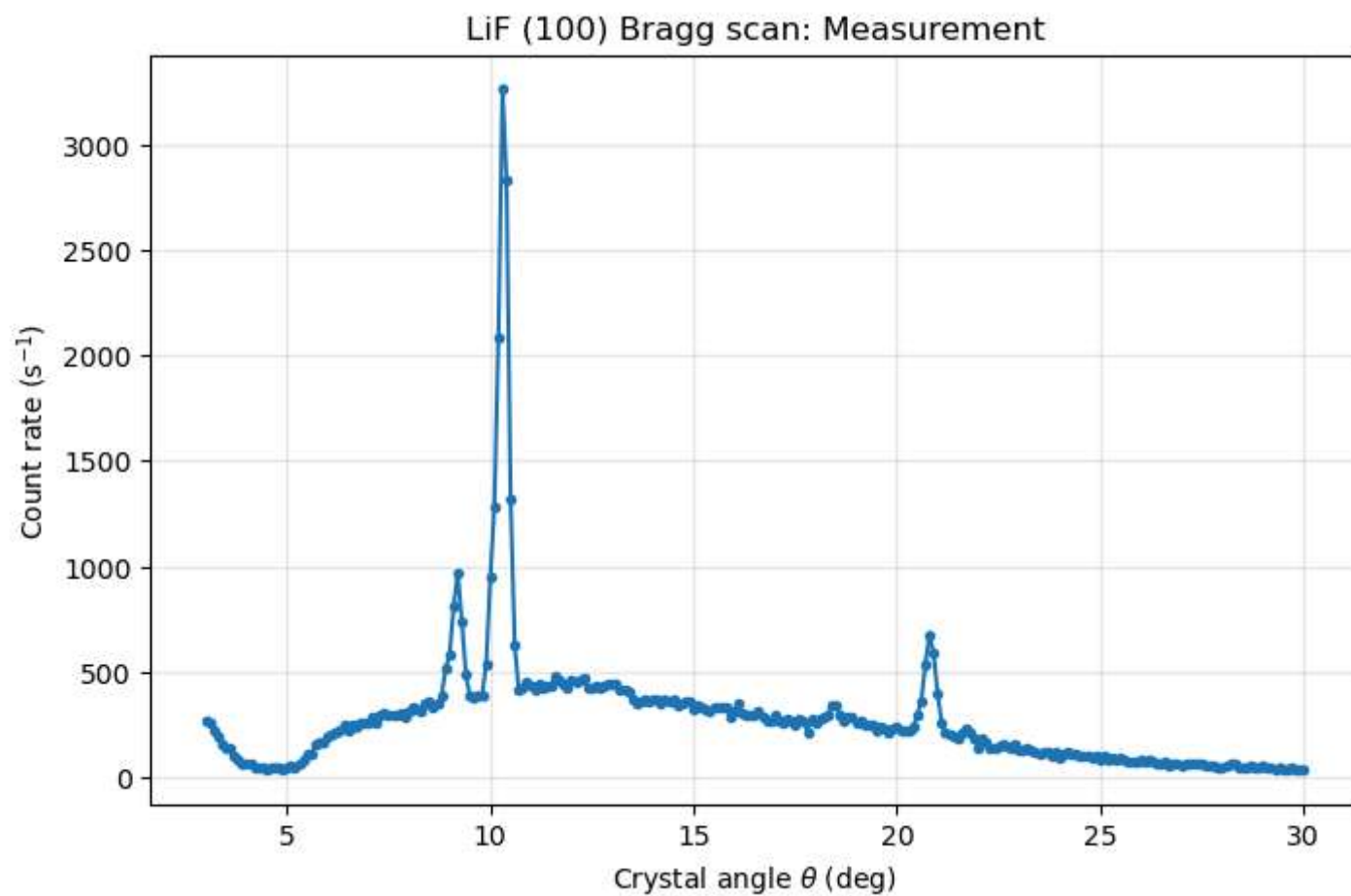
Data shape: (271, 2)

Don't need to drop first row, data numeric.

```
In [54]: plt.figure(figsize=(8,5))
plt.plot(LiF_100df["theta_deg"], LiF_100df["count_rate"], 'o-', markersize=3)
plt.xlabel(r"Crystal angle $\theta$ (deg)")
plt.ylabel("Count rate (s$^{-1}$)")
plt.title("LiF (100) Bragg scan: Measurement")
plt.grid(alpha=0.3)

# Save to specific location
plt.savefig(
    r"C:\Users\joshb\OneDrive\PHYSICS BSc UNIVERSITY of EXETER\BSc Physics Stage 2\Practical Physics II\practical_physics_II\Lab_2_Bragg_La
    dpi=300,
    bbox_inches="tight"
)

plt.show()
```



Graph Visualisation LiF (100) measurement:

```
In [56]: # Read into pandas DataFrame
KBr_100df = pd.read_csv(
    r"C:\Users\joshb\OneDrive\PHYSICS BSc UNIVERSITY of EXETER\BSc Physics Stage 2\Practical Physics II\practical_physics_II\Lab_2_Bragg_La
    sep="\t",
    encoding="latin1"
)

In [57]: # Rename columns
KBr_100df.columns = ["theta_deg", "count_rate"]

In [58]: display(KBr_100df.head())
print("Data shape:", KBr_100df.shape)
```

	theta_deg	count_rate
0	3.0	254
1	3.1	237
2	3.2	212
3	3.3	185
4	3.4	155

Data shape: (271, 2)

Don't need to drop first row, data numeric.

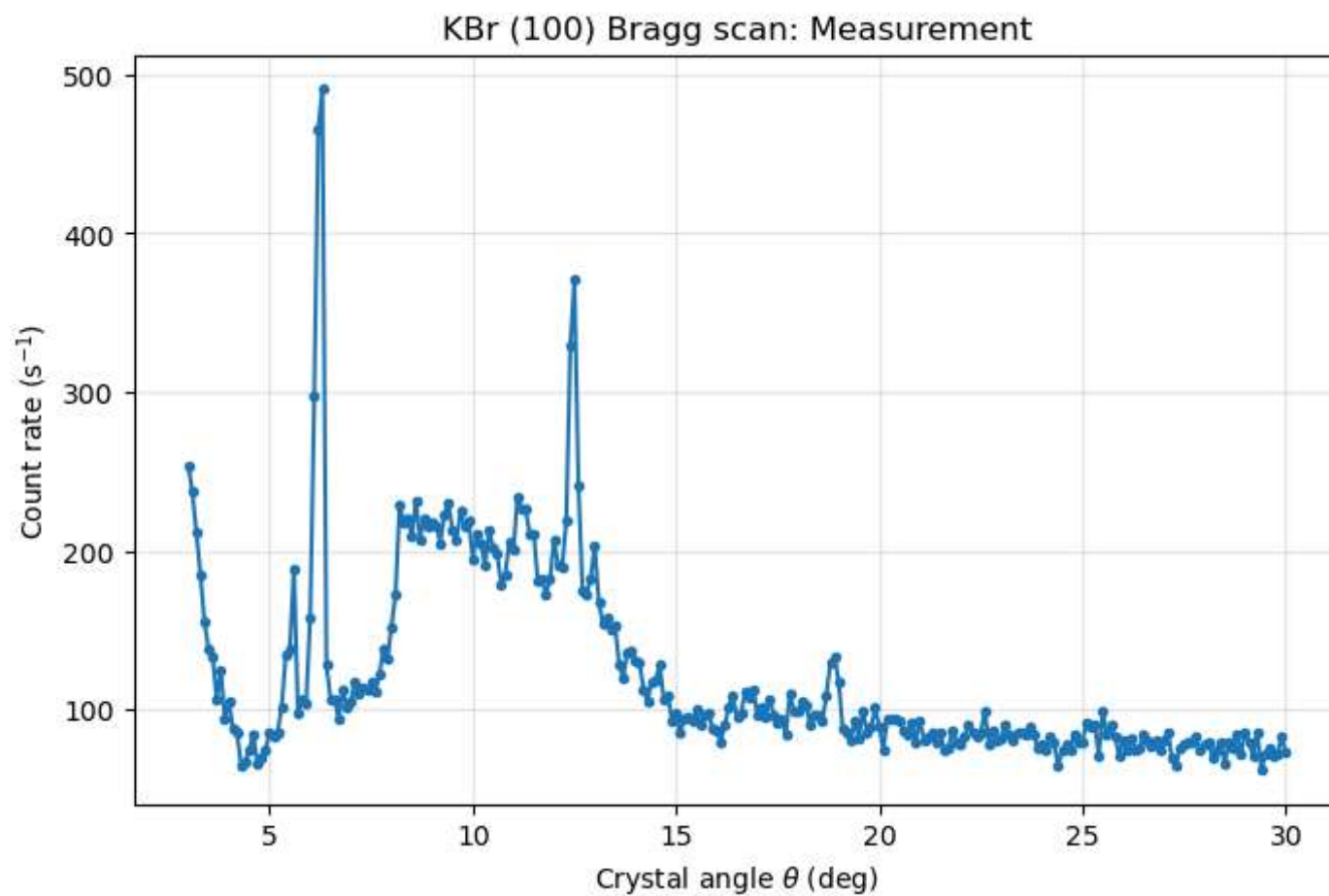
Error bars: NaCl (100) first measurement:

Appending pandas DataFrame to include columns for angular uncertainty and intensity uncertainty.

```
In [61]: plt.figure(figsize=(8,5))
plt.plot(KBr_100df["theta_deg"], KBr_100df["count_rate"], 'o-', markersize=3)
plt.xlabel(r"Crystal angle $\theta$ (deg)")
plt.ylabel("Count rate (s$^{-1}$)")
plt.title("KBr (100) Bragg scan: Measurement")
plt.grid(alpha=0.3)

# Save to specific location
plt.savefig(
    r"C:\Users\joshb\OneDrive\PHYSICS BSc UNIVERSITY of EXETER\BSc Physics Stage 2\Practical Physics II\practical_physics_II\Lab_2_Bragg_La
    dpi=300,
    bbox_inches="tight"
)

plt.show()
```



Angle uncertainty, count rate uncertainty and error bars

Calculating uncertainty values and adding error bars to graphs.

Angle uncertainty

Angle uncertainty is fixed, and given by the resolution of the X-ray unit.

Since step count is $\theta_{\text{step}} = 0.1^\circ$, then angular uncertainty is:

$$\Delta_{\text{angular}} = \pm 0.05^\circ$$

Count rate uncertainty

A Geiger-Müller tube can be modelled as a Poisson distribution when there are enough data points [1]. Thus uncertainties in the measured count rate can be assumed to follow Poisson statistics.

For a Poisson process with mean count N :

$$P(n) = \frac{N^n e^{-N}}{n!}$$

Thus variance is (see page * in lab book for derivation):

$$\text{variance} = \Delta^2 = N$$

Therefore:

$$\Delta_N = \sqrt{N}$$

Converting to count rate

N counts are recorded in time T . 'Count rate' is another way of saying intensity I over time period T .

Then:

$$I = \frac{N}{T} \quad \quad \Delta_I = \frac{\sqrt{N}}{T}$$

Gate time $T = 2\text{s}$.

Therefore:

$$\Delta_I = \sqrt{\frac{I}{2}}$$

Reference

[1] University of California, Davis, *Counting Statistics*.

<https://122.physics.ucdavis.edu/sites/default/files/files/Nuclear%20Decay/Counting%20Statistics%20122.pdf>

Accessed: 19/12/2025

Error Bars: NaCl (100) first reading

```
In [64]: # Angular uncertainty
NaCl_100df["delta_theta"] = 0.05
```

```
In [65]: # Intensity uncertainty computation
NaCl_100df["delta_intensity"] = np.sqrt(NaCl_100df["count_rate"] / 2)
```

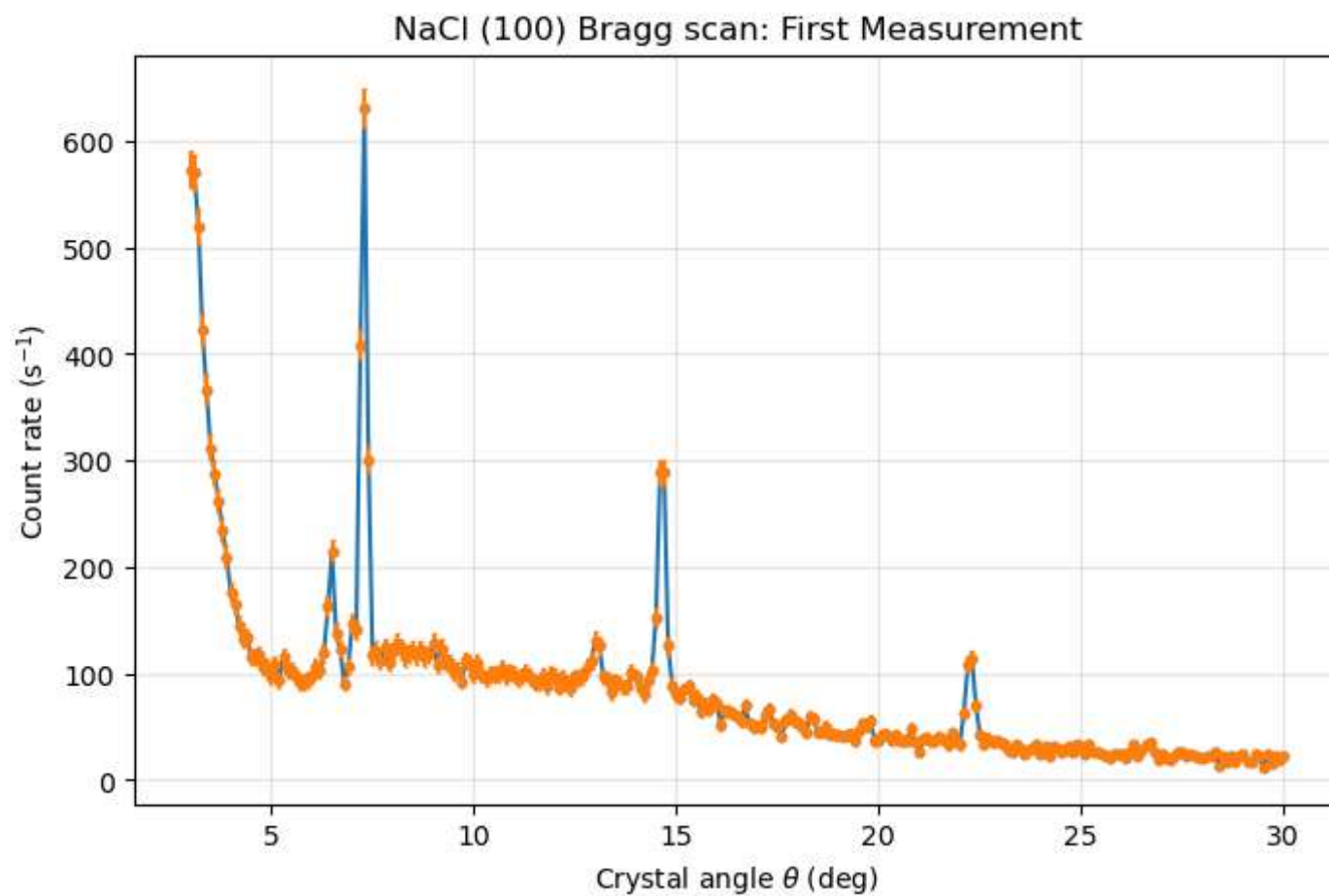
```
In [66]: display(NaCl_100df.head())
```

	theta_deg	count_rate	delta_theta	delta_intensity
0	3.0	573	0.05	16.926311
1	3.1	571	0.05	16.896745
2	3.2	520	0.05	16.124515
3	3.3	422	0.05	14.525839
4	3.4	367	0.05	13.546217

```
In [67]: plt.figure(figsize=(8,5))
plt.plot(NaCl_100df["theta_deg"], NaCl_100df["count_rate"], 'o-', markersize=3)
plt.xlabel(r"Crystal angle $\theta$ (deg)")
plt.ylabel("Count rate (s$^{-1}$)")
plt.title("NaCl (100) Bragg scan: First Measurement")
plt.grid(alpha=0.3)

plt.errorbar(
    NaCl_100df["theta_deg"],
    NaCl_100df["count_rate"],
    xerr=NaCl_100df["delta_theta"],
    yerr=NaCl_100df["delta_intensity"],
    fmt=".",
    capsize=1
)

plt.show()
```

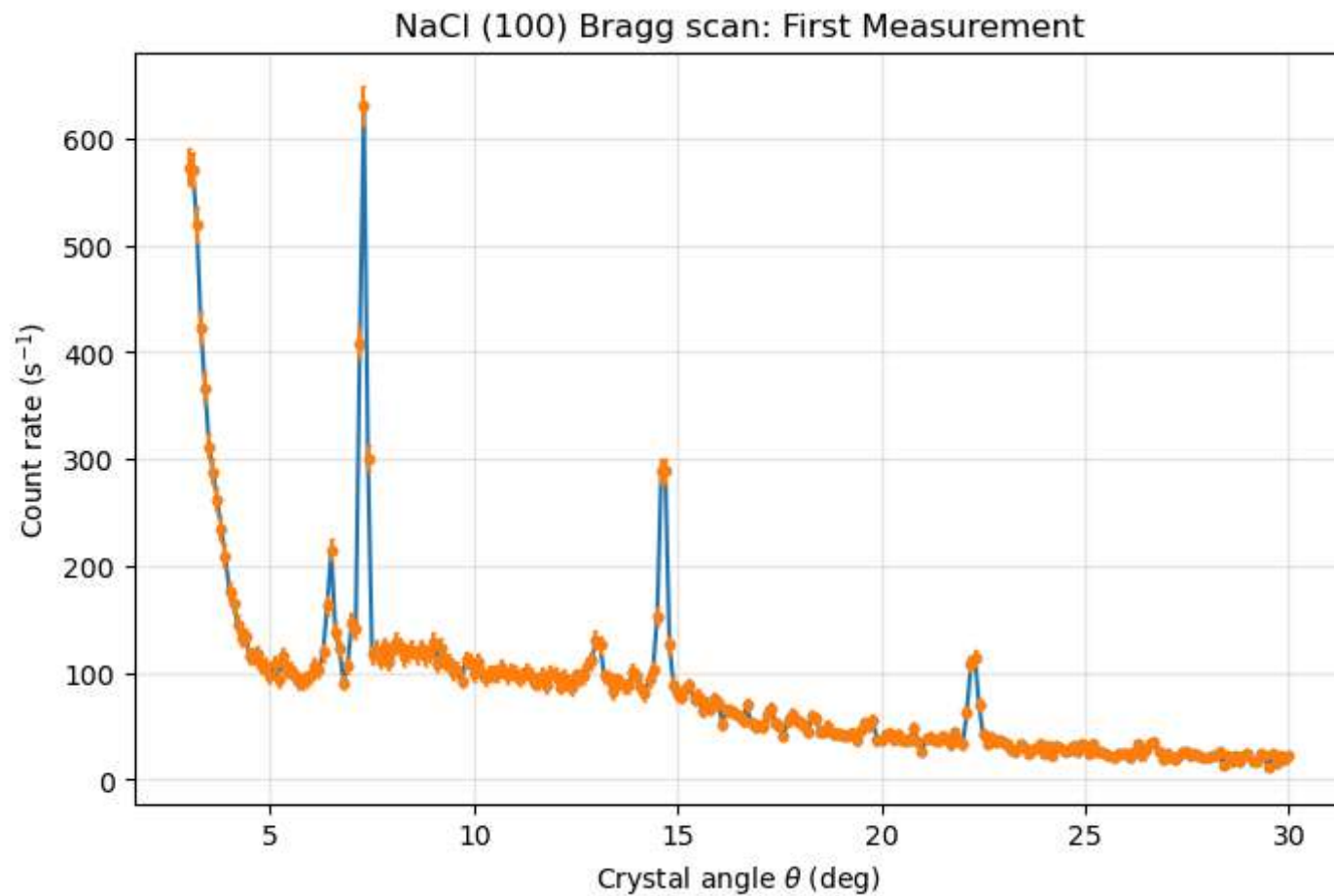


This is hard to read with the θ error bars plotted. Since $\Delta\theta$ is constant for all data points and significantly smaller than the angular range of the scan. Plotting angular error bars therefore adds no additional information and obscures the identification of Bragg peaks. The uncertainty in θ is instead included explicitly in the calculation of interplanar spacings

```
In [69]: plt.figure(figsize=(8,5))
plt.plot(NaCl_100df["theta_deg"], NaCl_100df["count_rate"], 'o-', markersize=3)
plt.xlabel(r"Crystal angle  $\theta$  (deg)")
plt.ylabel("Count rate (s-1)")
plt.title("NaCl (100) Bragg scan: First Measurement")
plt.grid(alpha=0.3)

plt.errorbar(
    NaCl_100df["theta_deg"],
    NaCl_100df["count_rate"],
    yerr=NaCl_100df["delta_intensity"],
    fmt=".",
```

```
capsize=1  
)  
  
plt.show()
```



This is still hard to read and not very useful. The uncertainty varies strongly with intensity and plotting error bars on every data point obscures the overall structure of the scan and the identification of Bragg peaks.

Since the primary purpose of these plots is qualitative peak identification rather than quantitative fitting, I am going to omit error bars for clarity. The relevant uncertainties were instead included explicitly in the numerical determination of Bragg angles and interplanar spacings.